NO.14 FAST IDLE/RUNS ON [SKYACTIV-D 1.5]

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14	FAST IDLE/RUNS ON
DESCRIPTION	• Engine speed continues at fast idle after warm-up. • Engine runs after ignition is switched off.
POSSIBLE CAUSE	• Improper engine oil level (close to or exceeds X mark) • PCM DTC is stored. • Erratic signal to PCM — Engine speed signal • Fuel injection system malfunction — Fuel leakage from fuel system — Common rail malfunction — Supply pump malfunction — Suction control valve malfunction — Fuel injector malfunction — Fuel pressure relief valve malfunction • Mechanical (engine) malfunction — Engine oil malfunction (oil working up or down)

Diagnostic Procedure

STEP	INSPECTION	RESULTS	ACTION
1	**INSPECT ENGINE OIL LEVEL** • Inspect the engine oil level. (See ENGINE OIL LEVEL INSPECTION [SKYACTIV-D 1.5].) • Is the engine oil level closed to or exceeds X mark?	Yes	Replace the engine oil and perform the “ENGINE OIL DATA RESET”. (See ENGINE OIL DATA RESET [SKYACTIV-D 1.5].)
		No	Go to the next step.
2	**VERIFY PCM DTC** • Retrieve PCM DTCs using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	Go to the next step.
3	**VERIFY CURRENT INPUT SIGNAL STATUS** **Caution** **• While performing this step, always operate the vehicle in a safe and lawful manner.** **• When the M-MDS is used to observe monitor system status while driving, be sure to have another technician with you, or record the data in the M-MDS using the PID/DATA MONITOR AND RECORD capturing function and inspect later.** • Access the following PIDs using the M-MDS: (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) — RPM — ARPMDES • Is the difference between the RPM monitor value and the ARPMDES monitor value within the specified value? (See ENGINE TUNE-UP [SKYACTIV-D 1.5].)	Yes	Inspect the related sensor and circuit. • If there is any malfunction: — Repair or replace the malfunctioning part according to the inspection results, then go to Step 6. • If there is no malfunction: — Go to the next step.
		No	Go to the next step.

STEP	INSPECTION	RESULTS	ACTION
4	**INSPECT FOR FUEL LEAKAGE FROM FUEL SYSTEM** • Visually inspect the following: — Fuel leakage from the fuel tank, fuel pump, hose, pipe, fuel injector, supply pump, common rail — Cracking and damage in fuel hose and pipe — Clamp installation condition for each hose and pipe — Fuel pipe securing condition due to deterioration such as rubber of clamp • Are all items normal?	Yes	Go to the next step.
		No	Repair or replace the malfunctioning part according to the inspection results, then go to Step 6.
5	**INSPECT FUEL INJECTION RELATED PARTS** • Inspect the following parts: — Common rail (See COMMON RAIL INSPECTION [SKYACTIV-D 1.5].) — Supply pump (See SUPPLY PUMP INSPECTION [SKYACTIV-D 1.5].) — Suction control valve (See SUCTION CONTROL VALVE INSPECTION [SKYACTIV-D 1.5].) — Fuel injector (See FUEL INJECTOR INSPECTION [SKYACTIV-D 1.5].) — Fuel pressure relief valve (See FUEL PRESSURE RELIEF VALVE INSPECTION [SKYACTIV-D 1.5].) • Are all items normal?	Yes	Inspect for the following engine internal parts: • Cylinder • Piston ring • Intake valve • Exhaust valve • Such as cylinder head gasket Repair or replace the malfunctioning part according to the inspection results, then go to the next step.
		No	Repair or replace the malfunctioning part according to the inspection results, then go to the next step.
6	Verify the test results. • If normal, return to the diagnostic index to service any additional symptoms. (See SYMPTOM DIAGNOSTIC INDEX [SKYACTIV-D 1.5].) • If a malfunction remains, inspect the related Service Information and perform the repair or diagnosis. — If the vehicle is repaired, troubleshooting is completed. — If the vehicle is not repaired or additional diagnostic information is not available, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)		